

Online Appendix for “*Strategic Concealment in Innovation Races*”

OA.1 Single-Firm Benchmark

Proposition OA.1.1. *Suppose that there is only one firm in the race, and it faces an exogenous termination rate λ . The optimal allocation policy is*

- (a) *to follow the R&D path if $\lambda < \lambda_*$;*
- (b) *to develop with the old technology if $\lambda > \lambda_*$.*

Proof of Proposition OA.1.1. Let $V_\lambda(N^i)$ denote the value function of Firm i in state N^i . If $N^i = 1$, the firm develops at rate λ_H until termination λ . The expected duration is $1/(\lambda_H + \lambda)$, and the success probability is $\lambda_H/(\lambda_H + \lambda)$. Thus, the continuation value is

$$V_\lambda(1) := \frac{\lambda_H}{\lambda_H + \lambda} \cdot \Pi - \frac{1}{\lambda_H + \lambda} \cdot c = \frac{\lambda_H \Pi - c}{\lambda_H + \lambda}. \quad (\text{OA.1.1})$$

If $N^i = 0$, the firm chooses allocation x to maximize the expected payoff. The corresponding Hamilton-Jacobi-Bellman (HJB) equation is:

$$0 = \max_{x \in [0,1]} \left\{ \lambda_L(1-x)(\Pi - V_\lambda(0)) + \mu x(V_\lambda(1) - V_\lambda(0)) + \lambda(0 - V_\lambda(0)) - c \right\}. \quad (\text{OA.1.2})$$

Solving for $V_\lambda(0)$ yields:

$$V_\lambda(0) = \max_{x \in [0,1]} \frac{\lambda_L(1-x)\Pi + \mu x V_\lambda(1) - c}{\lambda_L(1-x) + \mu x + \lambda}.$$

Differentiating the objective function with respect to x :

$$\frac{\lambda_L(\lambda \Pi + c)(\lambda_* - \lambda)}{(\lambda + \lambda_H)[\lambda + (1-x)\lambda_L + x\mu]^2}. \quad (\text{OA.1.3})$$

Therefore, the objective function is monotone in x and the sign of the first derivative is

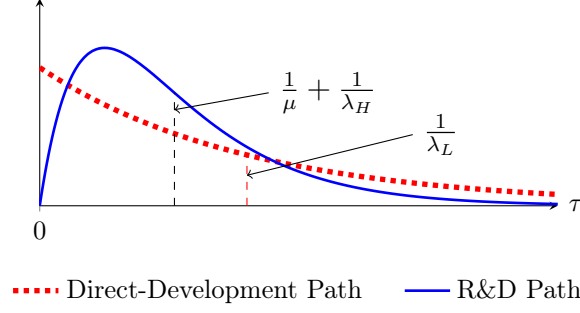


Figure OA.1: Probability distribution functions of a firm's development time

determined solely by the sign of $\lambda_\star - \lambda$. Thus, the optimal policy is to research ($\sigma_t = 1$) when $\lambda < \lambda_\star$, and to directly develop when $\lambda > \lambda_\star$. $\square$

To illustrate the intuition behind this proposition, Figure OA.1 shows the probability distributions of development times with the direct-development path (red dotted curve) and the R&D path (blue solid curve). The direct-development path is more likely to produce a successful product in a short time frame, as it requires only one breakthrough. In contrast, the R&D path produces a successful product after two breakthroughs and, although it has a shorter expected development time, is less likely to lead to quick development. Therefore, when facing a high termination rate, the firm prefers direct development to increase the probability of early success. Under a low termination rate, the firm prefers the R&D path to lower the expected development time.

OA.2 Non-stationary Extension: Private Progress

In the proof of Proposition 2, we restricted the strategy space to stationary (time-independent) allocations. In this section, we extend our analysis to the case of non-stationary allocations. We formulate the duration-minimization problem (Efficiency) and Firm A 's payoff-maximization problem (Equilibrium) over the unrestricted space of all measurable, time-dependent strategies $\sigma : \mathbb{R}_+ \rightarrow [0, 1]$. We then show that the same conclusion as in Proposition 2 holds.

OA.2.1 Efficiency (Social Planner's Problem)

We formulate the social planner's duration-minimization problem over all measurable allocations $\sigma_A, \sigma_B : \mathbb{R}_+ \rightarrow [0, 1]$. The planner's joint control is captured through the distribution of race phases, where $P_{jk}(t)$ denotes the probability that $N_t^A = j$ and $N_t^B = k$. Given the allocation pair $(\sigma_A(t), \sigma_B(t))$, the state dynamics are:

$$\dot{P}_{00} = -[2\lambda_L + (\mu - \lambda_L)(\sigma_A + \sigma_B)]P_{00}, \quad (\text{OA.2.1})$$

$$\dot{P}_{10} = \mu\sigma_A P_{00} - [\lambda_H + \lambda_L + (\mu - \lambda_L)\sigma_B]P_{10}, \quad (\text{OA.2.2})$$

$$\dot{P}_{01} = \mu\sigma_B P_{00} - [\lambda_L + (\mu - \lambda_L)\sigma_A + \lambda_H]P_{01}, \quad (\text{OA.2.3})$$

$$\dot{P}_{11} = \mu\sigma_A P_{01} + \mu\sigma_B P_{10} - 2\lambda_H P_{11}. \quad (\text{OA.2.4})$$

Since these phases are mutually exclusive, the race is ongoing at time t with probability $\sum_{jk} P_{jk}(t)$, and the planner minimizes the expected duration $J(\boldsymbol{\sigma}) = \int_0^\infty \sum_{jk} P_{jk}(t) dt$.

Proof of Proposition 2(a). We show via a verification argument that the constant policy $\sigma_A^* \equiv \sigma_B^* \equiv 1$ is the unique global minimizer of J .¹⁷

Let W_{jk}^* denote the stationary expected remaining durations from phase jk under $\boldsymbol{\sigma}^*$:

$$W_{11}^* = \frac{1}{2\lambda_H}, \quad W_{10}^* = W_{01}^* = \frac{1 + \mu W_{11}^*}{\lambda_H + \mu}, \quad W_{00}^* = \frac{1 + 2\mu W_{10}^*}{2\mu}, \quad (\text{OA.2.5})$$

and define the switching functions $\Phi_A(\mathbf{P}) = c_1 P_{00} + c_2 P_{01}$ and $\Phi_B(\mathbf{P}) = c_1 P_{00} + c_2 P_{10}$, with

$$c_1 := (\mu - \lambda_L)W_{00}^* - \mu W_{10}^*, \quad c_2 := (\mu - \lambda_L)W_{01}^* - \mu W_{11}^*.$$

Step 1: A global identity. Consider the linear function $V(\mathbf{P}) := \sum_{jk} W_{jk}^* P_{jk}$. Substituting (OA.2.1)–(OA.2.4), the derivative $\dot{V} = \sum_{jk} W_{jk}^* \dot{P}_{jk}$ is affine in (σ_A, σ_B) ; the recursions (OA.2.5) are exactly the conditions under which $\dot{V} = -\sum_{jk} P_{jk}$ at $\sigma_A = \sigma_B = 1$, and

¹⁷The standard sufficiency route via Arrow's theorem does not apply here: the maximized Hamiltonian is a pointwise maximum of functions affine in $\mathbf{P}$, hence convex, and whenever the fall-back condition holds the switching function changes sign on $\mathbb{R}_+^4$, rendering it strictly non-concave. The identity (OA.2.6) circumvents the need for concavity.

collecting the remaining terms gives

$$\dot{V}(\mathbf{P}(t)) = - \sum_{jk} P_{jk}(t) + (1 - \sigma_A(t))\Phi_A(\mathbf{P}(t)) + (1 - \sigma_B(t))\Phi_B(\mathbf{P}(t)).$$

Because the minimum exit rate $2\lambda_L > 0$, every admissible trajectory satisfies $\mathbf{P}(t) \rightarrow \mathbf{0}$, so $V(\mathbf{P}(t)) \rightarrow 0$. Integrating from $\mathbf{P}(0) = (1, 0, 0, 0)$, where $V(\mathbf{P}(0)) = W_{00}^*$, yields

$$J(\boldsymbol{\sigma}) = W_{00}^* + \int_0^\infty [(1 - \sigma_A(t))\Phi_A(\mathbf{P}(t)) + (1 - \sigma_B(t))\Phi_B(\mathbf{P}(t))] dt. \quad (\text{OA.2.6})$$

Step 2: Nonnegativity of the switching functions. Write $\Phi_A(\mathbf{P}(t)) = P_{00}(t)[c_1 + c_2 r(t)]$ with $r(t) := P_{01}(t)/P_{00}(t) \geq 0$. We first show $c_1 > 0$ unconditionally. Expanding,

$$c_1 = \frac{\mu\lambda_L\lambda_\star + \lambda_L\lambda_H(\lambda_\star - \mu)}{2\mu\lambda_H(\lambda_H + \mu)};$$

the numerator is positive from $\lambda_\star > \mu$. Hence $c_1 > 0$ for all admissible parameters.

The coefficient $c_2 = \frac{\lambda_L(\lambda_\star - \lambda_H)}{2\lambda_H(\lambda_H + \mu)}$ has the sign of the fall-back condition: $c_2 < 0$ if and only if $\lambda_\star < \lambda_H$.

If the fall-back condition fails, then $c_2 \geq 0$ and $\Phi_A(\mathbf{P}) = c_1 P_{00} + c_2 P_{01} \geq 0$ immediately.

If the fall-back condition holds, then $\lambda_H > \lambda_\star > \mu$ by Assumption (2.2), so $\bar{r} := \mu/(\lambda_H - \mu) > 0$ is well defined. Along any admissible trajectory, $r(0) = 0$ and $\dot{r} = \mu\sigma_B - r[(\lambda_H - \lambda_L) - (\mu - \lambda_L)\sigma_B]$; at $r = \bar{r}$, $\dot{r} = \frac{\mu(\lambda_H - \lambda_L)}{\lambda_H - \mu}(\sigma_B - 1) \leq 0$ for every $\sigma_B \leq 1$, so $r(t) \leq \bar{r}$ throughout. Since $c_2 < 0$, the affine map $r \mapsto c_1 + c_2 r$ attains its minimum on $[0, \bar{r}]$ at $\bar{r}$, where

$$c_1 + c_2 \bar{r} = \frac{\lambda_H \lambda_L (\lambda_\star - \mu)}{2\mu(\lambda_H^2 - \mu^2)} > 0$$

by Assumption (2.2). Hence $\Phi_A(\mathbf{P}(t)) \geq 0$. By symmetry, $\Phi_B(\mathbf{P}(t)) \geq 0$ in both cases.

Conclusion. Since $1 - \sigma_i(t) \geq 0$ and $\Phi_i(\mathbf{P}(t)) \geq 0$ for $i \in \{A, B\}$, the integrand in (OA.2.6) is nonnegative, so $J(\boldsymbol{\sigma}) \geq W_{00}^*$ for every admissible allocation. As $P_{00}(t) > 0$ for all finite t , the bracket $c_1 + c_2 r(t)$ is in fact strictly positive, so equality holds if and only if $\sigma_A(t) = \sigma_B(t) = 1$ for almost every t . Hence $\sigma_A^* \equiv \sigma_B^* \equiv 1$ is the unique global minimizer. $\square$

OA.2.2 Equilibrium (Firm A's Best Response)

We now formulate Firm A 's payoff-maximization problem over the unrestricted space of all measurable, time-dependent strategies $\sigma_A : \mathbb{R}_+ \rightarrow [0, 1]$. Suppose Firm B plays the constant allocation strategy $\sigma_B^*(t) = 1$. From Firm A 's perspective, Firm B 's hidden state probabilities $P_{B0}(t)$ and $P_{B1}(t)$ are exogenous and evolve as:

$$P_{B0}^*(t) = e^{-\mu t}, \quad P_{B1}^*(t) = \frac{\mu}{\lambda_H - \mu} (e^{-\mu t} - e^{-\lambda_H t}). \quad (\text{OA.2.7})$$

The probability that Firm B has not yet finished is $S_B^*(t) = P_{B0}^*(t) + P_{B1}^*(t)$.

If Firm A discovers the new technology at time t , its expected payoff relies on Firm B 's state at that exact instant. Let V_{10}^* and V_{11}^* be the stationary expected continuation values of racing in development against a rival in research or development, respectively:

$$V_{11}^* = \frac{\lambda_H \Pi - c}{2\lambda_H}, \quad V_{10}^* = \frac{\lambda_H \Pi + \mu V_{11}^* - c}{\lambda_H + \mu} = \frac{2\lambda_H + \mu}{\lambda_H + \mu} V_{11}^*.$$

Firm A 's conditionally expected continuation payoff upon discovery at time t is $U_A^*(t) = P_{B0}^*(t)V_{10}^* + P_{B1}^*(t)V_{11}^*$.

Let the state variable $R_A(t)$ denote the probability that Firm A has not yet finished the race, which evolves as $\dot{R}_A(t) = -[\lambda_L(1 - \sigma_A(t)) + \mu\sigma_A(t)]R_A(t)$. Firm A 's objective is to maximize:

$$\max_{\sigma_A} \int_0^\infty R_A(t) [\lambda_L(1 - \sigma_A(t))S_B^*(t)\Pi + \mu\sigma_A(t)U_A^*(t) - S_B^*(t)c] dt. \quad (\text{OA.2.8})$$

Proof of Proposition 2 (b). We again apply Arrow's sufficiency condition. Let $\eta_A(t)$ be the co-state variable associated with $R_A(t)$. The Hamiltonian is:

$$\begin{aligned} H(\sigma_A(t), R_A(t), \eta_A(t)) &= R_A(t) [\lambda_L(1 - \sigma_A(t))S_B^*(t)\Pi + \mu\sigma_A(t)U_A^*(t) - S_B^*(t)c] \\ &\quad - \eta_A(t)R_A(t) [\lambda_L + (\mu - \lambda_L)\sigma_A(t)]. \end{aligned} \quad (\text{OA.2.9})$$

We construct the candidate optimal paths $(\eta_A^*(t), R_A^*(t))$ generated by the constant policy

$\sigma_A^*(t) = 1$. The state trajectory is $R_A^*(t) = e^{-\mu t}$. Let the co-state variable be Firm A 's conditionally expected continuation value under the constant profile $(1, 1)$, which is exactly $\eta_A^*(t) = P_{B0}^*(t)V_{00}^* + P_{B1}^*(t)V_{01}^*$, where

$$V_{01}^* = \frac{\mu V_{11}^* - c}{\lambda_H + \mu}, \quad V_{00}^* = \frac{\mu V_{10}^* + \mu V_{01}^* - c}{2\mu} = \frac{1}{2}(V_{10}^* + V_{01}^*) - \frac{c}{2\mu}.$$

Arrow's sufficiency condition requires verifying four properties:

1. **Maximum Principle:** For all $t \geq 0$, the policy $\sigma_A^* = 1$ maximizes the Hamiltonian. Grouping the terms in the Hamiltonian containing $\sigma_A(t)$ and substituting $\eta_A^*(t)$, we obtain Firm A 's switching function:

$$\begin{aligned} \Phi_A(t) &= -\lambda_L S_B^*(t)\Pi + \mu U_A^*(t) - (\mu - \lambda_L)\eta_A^*(t) \\ &= P_{B0}^*(t) \underbrace{\left[\mu V_{10}^* - \lambda_L \Pi - (\mu - \lambda_L)V_{00}^* \right]}_{\Delta_{00}^*} + P_{B1}^*(t) \underbrace{\left[\mu V_{11}^* - \lambda_L \Pi - (\mu - \lambda_L)V_{01}^* \right]}_{\Delta_{01}^*}, \end{aligned} \tag{OA.2.10}$$

where $\sigma_A^* = 1$ maximizes the Hamiltonian if and only if $\Phi_A(t) \geq 0$.

By substituting (OA.2.7) into (OA.2.10), we can derive the exact algebraic form:

$$\Phi_A(t) = e^{-\mu t} \left[\Delta_{00}^* + \frac{\mu}{\lambda_H - \mu} (1 - e^{-(\lambda_H - \mu)t}) \Delta_{01}^* \right].$$

With some algebra, we can derive that

$$\begin{aligned} \Phi_A(t) &= \frac{\lambda_L(\lambda_\star - \mu)}{2\mu} \left(\frac{\mu}{\lambda_H + \mu} \Pi + \frac{c}{\lambda_H} \right) e^{-\mu t} \\ &\quad + \frac{\lambda_L \mu}{2(\lambda_H + \mu)} \left[\frac{\lambda_\star - \mu}{\lambda_H - \mu} (1 - e^{-(\lambda_H - \mu)t} + e^{-(\lambda_H - \mu)t}) \right] \left(\Pi + \frac{c}{\lambda_H} \right) e^{-\mu t}. \end{aligned}$$

Since $\lambda_\star > \mu$ and $\lambda_H > \mu$, $\Phi_A(t) > 0$ for all $t \geq 0$. Therefore, $\sigma_A^* = 1$ maximizes the Hamiltonian point-wise.

2. **Evolution of the co-state:** The co-state must satisfy $\dot{\eta}_A^*(t) = -\frac{\partial H}{\partial R_A}$ evaluated at

$\sigma_A^* = 1$. This yields the differential equation:

$$\dot{\eta}_A^*(t) = \mu\eta_A^*(t) - [\mu U_A^*(t) - S_B^*(t)c].$$

Note that $\dot{P}_{B0}^*(t) = -\mu P_{B0}^*(t)$ and $\dot{P}_{B1}^*(t) = \mu P_{B0}^*(t) - \lambda_H P_{B1}^*(t)$. Using these evolutions and the definitions of V_{00}^* and V_{01}^* , we can verify that the above equation holds.

3. **Transversality Condition:** We must show $\lim_{t \rightarrow \infty} \eta_A^*(t)(R_A^*(t) - R_A(t)) \leq 0$ for any feasible trajectory $R_A(t)$. For any admissible policy, the probabilities $R_A(t)$ and $R_A^*(t)$ are bounded in $[0, 1]$ and decay to 0 as $t \rightarrow \infty$ due to the minimum baseline hazard rate $\lambda_L > 0$. Because $\eta_A^*(t)$ represents bounded expected continuation payoffs, the limit evaluates exactly to 0, satisfying the condition.

4. **Concavity:** The maximized Hamiltonian

$$\hat{H}(R_A(t), \eta_A(t)) = \max_{\sigma_A(t) \in [0, 1]} H(\sigma_A(t), R_A(t), \eta_A(t))$$

must be concave in $R_A(t)$. As explicitly shown in (OA.2.9), the Hamiltonian is strictly linear in the state variable $R_A(t)$, thereby satisfying weak concavity.

Because all four conditions hold, Arrow's sufficiency theorem guarantees that the constant policy $\sigma_A^*(t) = 1$ is Firm A 's global best response among all measurable, time-dependent allocation strategies. Symmetrically, Firm B 's global best response to Firm A 's research strategy is also the research strategy. Therefore, the profile where both firms research constitutes an equilibrium. $\square$

OA.2.3 Optimality of Constant Resource Allocation Strategies

In the main text (Proof of Theorem 1), we restricted attention to constant resource allocation strategies. In this appendix, we use optimal control theory to rigorously prove that if a constant allocation strategy maximizes a firm's stationary expected payoff, it is a global best response among all measurable, time-dependent strategies.

Consider the state where neither firm has discovered the new technology. Suppose Firm B uses a constant allocation strategy $b \in [0, 1]$. Firm A employs a potentially time-dependent, measurable allocation policy $\sigma : \mathbb{R}_+ \rightarrow [0, 1]$.

Let V_A and V_B denote Firm A 's continuation payoffs if Firm A or Firm B discovers the new technology first, respectively. Under the candidate equilibrium, both firms immediately patent upon discovery, locking in the licensing subgame payoffs: $V_A = V_{11} + l^*$ and $V_B = V_{11} - l^*$. The flow rate at which Firm A develops the old technology is $\lambda_L(1 - \sigma_t)$, and the rates at which Firm A and Firm B discover the new technology are $\mu\sigma_t$ and μb , respectively.

Let $\Sigma_t = \int_0^t \sigma_s ds$ be Firm A 's cumulative research effort. The probability that the race is still ongoing at time t (the survival function) is:

$$r_t = \exp \left(- \int_0^t \{ \mu(\sigma_s + b) + \lambda_L(2 - \sigma_s - b) \} ds \right). \quad (\text{OA.2.11})$$

Equivalently, the state variable r_t evolves according to the differential equation:

$$\dot{r}_t = - \{ \mu(\sigma_t + b) + \lambda_L(2 - \sigma_t - b) \} \cdot r_t, \quad \text{with } r_0 = 1. \quad (\text{OA.2.12})$$

Firm A 's objective is to choose a policy σ_t to maximize its expected payoff from the race:

$$J(\sigma) = \int_0^\infty \{ \lambda_L(1 - \sigma_t)\Pi + \mu\sigma_t V_A + \mu b V_B - c \} \cdot r_t dt. \quad (\text{OA.2.13})$$

To link this dynamic problem to the stationary formulation in the main text, let $N(x, b)$ denote the flow payoff term inside the brackets of (OA.2.13) evaluated at a constant x , and let $D(x, b)$ denote the aggregate hazard rate term inside the brackets of (OA.2.12). If Firm A also plays a constant strategy $\sigma_t = x$, the expected payoff reduces exactly to the stationary form $U(x) = \frac{N(x, b)}{D(x, b)}$.

Lemma OA.2.1. *Suppose there exists $x^* \in [0, 1]$ such that $x^* \in \arg \max_{x \in [0, 1]} U(x)$. Then the constant policy $\sigma_t^* = x^*$ for all $t \geq 0$ is a global maximum of the optimal control problem defined by (OA.2.13) and (OA.2.12).*

Proof of Lemma OA.2.1. We apply Arrow's sufficiency condition for infinite-horizon optimal control problems (e.g., Seierstad and Sydsaeter, 1987, Theorem 3.14). Let η_t be the co-state

variable associated with r_t . The Hamiltonian is:

$$\begin{aligned} H(\sigma_t, r_t, \eta_t) &= N(\sigma_t, b) \cdot r_t - \eta_t \cdot D(\sigma_t, b) \cdot r_t \\ &= \{N(\sigma_t, b) - \eta_t D(\sigma_t, b)\} \cdot r_t. \end{aligned} \quad (\text{OA.2.14})$$

We construct the candidate optimal paths (η_t^*, r_t^*) generated by the constant policy $\sigma_t^* = x^*$. Let the co-state variable be constant: $\eta_t^* = U(x^*)$ for all $t \geq 0$. The state trajectory under x^* is $r_t^* = e^{-D(x^*, b)t}$.

Arrow's sufficiency condition requires verifying four properties:

1. **Maximum Principle:** For all $t \geq 0$, $\sigma_t^* = x^*$ maximizes the Hamiltonian. Substituting $\eta_t^* = U(x^*)$, we have:

$$H(\sigma_t, r_t^*, \eta_t^*) = \{N(\sigma_t, b) - U(x^*)D(\sigma_t, b)\} \cdot r_t^*. \quad (\text{OA.2.15})$$

Because x^* maximizes $U(x) = N(x, b)/D(x, b)$, we know that for any $\sigma_t \in [0, 1]$, $U(x^*) \geq N(\sigma_t, b)/D(\sigma_t, b)$. Since $D(\sigma_t, b) > 0$, this implies $N(\sigma_t, b) - U(x^*)D(\sigma_t, b) \leq 0$. Thus, $H(\sigma_t, r_t^*, \eta_t^*) \leq 0$ for all admissible σ_t . Furthermore, evaluating at $\sigma_t = x^*$ yields exactly 0. Therefore, $\sigma_t^* = x^*$ maximizes the Hamiltonian point-wise.

2. **Evolution of the co-state:** The co-state must satisfy $\dot{\eta}_t^* = -\frac{\partial H}{\partial r_t}(x^*, r_t^*, \eta_t^*)$. Because $H(x^*, r_t^*, \eta_t^*) = 0$ for all r_t^* , its partial derivative with respect to r_t is 0. This is perfectly consistent with our construction that $\dot{\eta}_t^* = 0$.
3. **Transversality Condition:** We must show $\lim_{t \rightarrow \infty} \eta_t^*(r_t^* - r_t) \leq 0$ for any feasible state trajectory r_t . Note that for any admissible policy σ , $r_t \geq 0$ and $r_t^* \geq 0$. Furthermore, because $D(\sigma_t, b) \geq 2\lambda_L > 0$, both r_t and r_t^* converge strictly to 0 as $t \rightarrow \infty$. Since $\eta_t^* = U(x^*)$ is a finite constant, the limit evaluates exactly to 0, satisfying the condition.
4. **Concavity:** The maximized Hamiltonian $\hat{H}(r_t, \eta_t) = \max_{\sigma_t \in [0, 1]} H(\sigma_t, r_t, \eta_t)$ must be concave in r_t . As seen in (OA.2.14), the Hamiltonian is strictly linear in r_t , satisfying weak concavity.

Because all four conditions hold, Arrow's sufficiency theorem guarantees that the constant policy $\sigma_t^* = x^*$ is a global best response among all measurable allocation strategies. $\square$

OA.3 Proofs for Section 5

In this section, we provide the proofs for the results presented in Section 5. The exposition is organized as follows: Section OA.3.1 formalizes the evolution of firms' beliefs and develops a recursive formulation of the patenting optimization problem. Section OA.3.2 proves that any equilibrium patenting strategy must take the form of a gradual patenting strategy (Proposition 4). Building on this structure, Section OA.3.3 contains the proof of the main equilibrium characterization (Theorem 2). Proofs of all intermediate lemmas are relegated to Section OA.3.4.

OA.3.1 Belief Evolution and Recursive Formulation

Belief Evolution Suppose that a firm employs the patent filing policy $G = \{G_s\}_{s \geq 0}$. Let $\mathbf{p}_t$ denote the probability that the firm has the new technology at time t (conditional on the firm having neither developed the product nor patented the new technology). The following lemma characterizes the evolution of $\mathbf{p}$.

Lemma OA.3.1. *Suppose that $\sigma = \mathbf{1}$ and the patent filing strategy is G . Then, $\mathbf{p}$ evolves as follows:*

$$d\mathbf{p}_t = (1 - \mathbf{p}_t) \cdot [(\mu - \lambda_H \mathbf{p}_t)dt - d\mathcal{P}_t], \quad (\text{OA.3.1})$$

where $d\mathcal{P}_t$ represents the patenting intensity—the instantaneous probability that the firm files a patent at time t —given by

$$d\mathcal{P}_t = \mu(1 - \mathbf{p}_t) \left[G_t(t)dt + \int_0^t e^{-(\lambda_H - \mu)(t-s)} dG_s(t)ds \right].^{18} \quad (\text{OA.3.2})$$

¹⁸We interpret $dG_s(t)$ in the Lebesgue-Stieltjes sense. If G_s is continuous at t , then $dG_s(t) = g_s(t)dt$, where $g_s(t)$ denotes the right-derivative of G_s . If G_s is discontinuous at t , then $dG_s(t)$ represents the probability mass of the atom, $G_s(t) - G_s(t^-)$.

Recursive Formulation Now suppose that Firm A discovers the new technology at time τ , chooses the patent filing policy $\tilde{G}_\tau$ —a CDF over $[\tau, \infty]$ —and Firm B uses the patent filing policy $G := \{G_s\}_{s \geq 0}$.

Given the resource allocation strategy $\sigma = \mathbf{1}$, the probability that Firm B has neither filed the patent nor developed the product by time t is $e^{-\mu t}/(1 - \mathbf{p}_t)$.¹⁹ Define

$$K_t := \frac{e^{-(\lambda_H + \mu)t}}{1 - \mathbf{p}_t}. \quad (\text{OA.3.3})$$

Then, the term K_t/K_τ represents the probability that Firm A has not developed the product and Firm B has neither developed the product nor filed a patent by time t .

We first consider the full concealment benchmark: $\tilde{G}_\tau(t) = 0$ for all $t \geq \tau$. Let V_t^n denote the expected payoff in this case. The following lemma characterizes V_t^n .

Lemma OA.3.2. *Suppose Firm B uses the patent filing policy G . Then, Firm A's expected payoff from the full concealment policy is:*

$$V_\tau^n := \frac{1}{K_\tau} \int_\tau^\infty d\mathcal{F}_t \quad (\text{OA.3.4})$$

where

$$d\mathcal{F}_t = (\lambda_H \Pi - c) \cdot K_t dt + (V_{11} - (1 - \beta) \cdot l^*) \cdot K_t d\mathcal{P}_t. \quad (\text{OA.3.5})$$

Additionally, for all $t \geq 0$, $|V_t^n| < \infty$.

Define $\Phi(t) := \{(V_{11} + (1 - \beta \mathbf{p}_t) \cdot l^*) - V_t^n\} \cdot K_t$ as the *discounted excess payoff* of patenting over concealment—the difference in discounted expected payoffs between immediate patenting and full concealment. The following lemma shows that we can write Firm A's expected payoff (given Firm B's policy) using $\Phi(t)$.

Lemma OA.3.3. *Suppose Firm A discovers the new technology at time τ and employs the patent filing policy $\tilde{G}_\tau$, and Firm B uses the patent filing policy G . Then, Firm A's expected*

¹⁹Note that $\Pr(\text{Firm B has not discovered the new technology}) = e^{-\mu t}$ and

$$1 - \mathbf{p}_t = \frac{\Pr(\text{Firm B has not discovered the new technology})}{\Pr(\text{Firm B has neither filed a patent nor developed the product})}.$$

payoff at time τ can be written as follows:

$$V_\tau(\tilde{G}_\tau, G) = V_\tau^n + \frac{1}{K_\tau} \left[\tilde{G}_\tau(\tau) \cdot \Phi(\tau) + \int_\tau^\infty \Phi(t) d\tilde{G}_\tau(t) \right]. \quad (\text{OA.3.6})$$

This lemma demonstrates a powerful property of the firm's optimization problem: the expected payoff from any arbitrary patenting strategy can be linearly decomposed. It is exactly equal to the baseline payoff of never patenting, V_τ^n , plus the expected incremental value of disclosure. This incremental value is captured by integrating the discounted excess payoff $\Phi(t)$ over the firm's chosen filing distribution $\tilde{G}_\tau$. The first term inside the brackets accounts for the immediate payoff jump if the firm files a discrete probability mass exactly at the moment of discovery, while the integral captures the expected excess benefit of any delayed filings.

The following lemma shows that the support of the optimal patenting policy should be concentrated on the times that maximize the discounted excess payoff.

Lemma OA.3.4. *Let G_τ^* be an optimal patenting policy for a firm that discovered the new technology at time τ : $G_\tau^* \in \arg \max_{\tilde{G}_\tau} V_\tau(\tilde{G}_\tau, G)$.*

- (a) *If $\sup_{t \geq \tau} \Phi(t) < 0$, then full concealment is optimal: $G_\tau^*(t) = 0$ for all $t \geq \tau$.*
- (b) *If T is in the support of dG_τ^* , then T must maximize the discounted excess payoff:*

$$T \in \arg \max_{t \geq \tau} \Phi(t) \quad \text{and} \quad \Phi(T) \geq 0.$$

This result is a direct consequence of the linear payoff decomposition established in Lemma OA.3.3. Because the firm acts to maximize an expectation over the measure $d\tilde{G}_\tau$, its optimal strategy must allocate probability mass exclusively to the moments in time that yield the highest possible excess payoff. If $\Phi(t)$ remains strictly negative for all $t \geq \tau$, then any act of patenting strictly reduces the firm's expected payoff relative to the baseline, making full concealment the uniquely optimal choice. Conversely, if the firm does choose to patent, it will only do so at times T that globally maximize the excess payoff, provided that this maximum is at least zero. This dramatically simplifies our equilibrium analysis: identifying

the optimal timing of patenting is equivalent to tracking the trajectory of $\Phi(t)$ and locating its peaks.

OA.3.2 Gradual Patenting Strategy: Proof of Proposition 4

Now we show that every equilibrium patenting strategy takes the form of a gradual patenting strategy (Proposition 4). We first establish the differential evolution of the excess payoff.

Lemma OA.3.5. *In any symmetric research equilibrium, the discounted excess payoff Φ evolves as follows:*

$$\frac{d\Phi(t)}{K_t V_{11}} = (1 - \mathbf{p}_t) \lambda_H dt - \rho \cdot \{(\lambda_H(1 + \mathbf{p}_t - 2\beta \mathbf{p}_t) + \beta \mu(1 - \mathbf{p}_t)) dt + 2(1 - \beta) d\mathcal{P}_t\}. \quad (\text{OA.3.7})$$

This lemma decomposes the instantaneous change in the excess payoff of patenting. The evolution is driven by the natural progression of the race and the rival's patenting intensity, $d\mathcal{P}_t$. Crucially, the term scaled by ρ demonstrates that a higher relative value of licensing exerts stronger downward pressure on Φ —making immediate patenting more attractive than delay when ρ is high. Furthermore, the equation reveals that any active patenting by the rival ($d\mathcal{P}_t > 0$) also decreases a firm's own excess payoff—further pressing firms to patent earlier.

Lemma OA.3.6. *In any symmetric research equilibrium, Φ has no discrete jump.*

This lemma rules out mass points in the patenting timing. If firms were expected to file a patent at a specific, predictable instant, the simultaneous surge in disclosure would discontinuously destroy the value of exclusivity. Anticipating this sudden drop in value, a firm would strictly prefer to preempt the mass point by filing slightly earlier, thereby smoothing out the distribution of patent filings over time.

Lemma OA.3.7. *In any symmetric research equilibrium, if $\Phi(T) = \Phi(T')$ for some $T < T'$, then for any $S \in (T, T')$, we have $\Phi(S) \geq \Phi(T) = \Phi(T')$.*

This result implies that the excess payoff trajectory cannot have “valleys” or be U-shaped between optimal patenting times. During periods where no patenting occurs, a firm's belief $\mathbf{p}_t$ that the rival has the technology strictly increases.

Proof of Proposition 4. By Lemma OA.3.4, the support of any optimal patenting policy G_τ^* must be contained strictly within the set of times that maximize the discounted excess payoff $\Phi(t)$ over $t \geq \tau$.

Lemma OA.3.7 establishes that the set of global maximizers of $\Phi(t)$ on $\mathbb{R}_+$ cannot contain any gaps. Therefore, the set of times that maximize $\Phi(t)$ must form a single connected interval, which we denote as $[T_1, T_2]$ with $0 \leq T_1 \leq T_2 \leq \infty$. Additionally, Φ must increase for $t < T_1$ and decrease for $t > T_2$.

This connected interval perfectly dictates the three phases of the firm's patenting behavior:

- (i) **Silent Phase:** For any $t < T_1$, $\Phi(t)$ has not yet reached its maximum. Thus, for any discovery at $\tau < T_1$, the optimal policy assigns zero probability mass to filing before T_1 . Therefore, $G_\tau(t) = 0$ for all $t < T_1$.
- (ii) **Mixing Phase:** For $t \in [T_1, T_2]$, $\Phi(t)$ is constant at its global maximum. If $T_1 < T_2$, the firm is indifferent regarding exactly when to patent within this interval, allowing for a continuous mixed strategy ($0 < G_\tau(t) < 1$). Lemma OA.3.6 ensures there are no discrete jumps (atoms) in this mixing distribution. Because $\Phi(t)$ strictly decreases after T_2 , the optimal window closes, meaning the firm must have patented with certainty by the end of the interval: $G_\tau(T_2) = 1$.
- (iii) **Immediate Phase:** For $t > T_2$, $\Phi(t)$ is strictly decreasing. If a firm discovers the new technology at a time $\tau \geq T_2$, the maximum of $\Phi(t)$ on the restricted domain $[\tau, \infty)$ occurs exactly at the boundary $t = \tau$. Thus, the firm's unique best response is to patent immediately upon discovery, yielding $G_\tau(\tau) = 1$ (and generally $G_t(t) = 1$ for all $t \geq T_2$).

Because any symmetric equilibrium patenting strategy must follow this exact sequential structure across the interval $[T_1, T_2]$, it is, by definition, a gradual patenting strategy. $\square$

OA.3.3 Proof of Theorem 2

In this subsection, we present the formal proof of Theorem 2. We begin by establishing a foundational lemma that maps the structural phases of any gradual patenting strategy—defined by its phase transitions (T_1, T_2) —to precise parametric conditions. By evaluating the trajectory of the discounted excess payoff, this lemma provides the necessary conditions that systematically rule out impossible strategy profiles across the parameter space.

Lemma OA.3.8. *Suppose that a gradual patenting strategy with thresholds (T_1, T_2) constitutes a symmetric research equilibrium. Then, the following statements hold:*

- (a) if $T_1 = \infty$, then $\rho \leq \rho_D$;
- (b) if $T_1 < \infty$ and $T_2 = \infty$, then (i) $\rho \in (\rho_D, \rho_M]$, (ii) for all $t \geq T_1$, $\Phi(t) = 0$ and

$$\mathbf{p}_t = \mathbf{p}^* := 1 - \frac{2\mu(1 - \beta)\rho}{\lambda_H(1 - \rho) - \mu\beta\rho};$$

- (c) if $T_2 < \infty$, then $\rho > \rho_M$.

Proof of Lemma OA.3.8. From Lemma OA.3.5, the differential evolution of the discounted excess payoff is given by:

$$d\Phi = K_t V_{11} [(1 - \mathbf{p}_t)\lambda_H dt - \rho \cdot \{(\lambda_H(1 + \mathbf{p}_t - 2\beta\mathbf{p}_t) + \beta\mu(1 - \mathbf{p}_t)) dt + 2(1 - \beta)d\mathcal{P}_t\}].$$

By the definition of the gradual patenting strategy, $\Phi(t)$ attains its global maximum on the interval $[T_1, T_2]$.

(a) Case $T_1 = \infty$ (Full Concealment): Firms never patent, so $d\mathcal{P}_t = 0$ for all $t \geq 0$. For full concealment to be optimal, we must have $\Phi(t) \leq 0$ for all $t \geq 0$. Under $d\mathcal{P}_t = 0$, we have $M_t = e^{-\mu t}$ and $L_t = \frac{\mu}{\lambda_H - \mu}(e^{-\mu t} - e^{-\lambda_H t})$. The belief $\mathbf{p}_t = \frac{L_t}{M_t + L_t}$ simplifies to:

$$\mathbf{p}_t = \frac{\mu(e^{-\mu t} - e^{-\lambda_H t})}{\lambda_H e^{-\mu t} - \mu e^{-\lambda_H t}},$$

which is strictly increasing and converges to μ/λ_H as $t \rightarrow \infty$. Calculating the continuation value $V_t^n = \frac{1}{K_t} \int_t^\infty 2\lambda_H V_{11} K_s ds$ yields $V_t^n = V_{11}[1 + \frac{\lambda_H}{\lambda_H + \mu}(1 - \mathbf{p}_t)]$. Substituting this into the

condition $\Phi(t) \leq 0$ gives:

$$\Phi(t) = (1 - \beta \mathbf{p}_t) V_{11} K_t \left[\rho - \frac{\lambda_H}{\lambda_H + \mu} \cdot \frac{1 - \mathbf{p}_t}{1 - \beta \mathbf{p}_t} \right] \leq 0.$$

Since this must hold for all $t \geq 0$, and the right-hand bracket is strictly decreasing in $\mathbf{p}_t$, the inequality must hold at the limit $t \rightarrow \infty$ where $\mathbf{p}_t \rightarrow \mu/\lambda_H$. This limit evaluates exactly to $\rho \leq \rho_D$.

(b) Case $T_1 < \infty$ and $T_2 = \infty$ (Delayed Mixing): The mixing phase lasts indefinitely, meaning $\Phi(t)$ is constant at its maximum for all $t \geq T_1$. Thus, $d\Phi(t) = 0$ for $t \geq T_1$. Setting $d\Phi = 0$ in the evolution equation yields:

$$A := \frac{\lambda_H - (\lambda_H + \beta\mu)\rho}{2(1 - \beta)\rho} = \frac{\lambda_H \mathbf{p}_t}{1 - \mathbf{p}_t} + \frac{d\mathcal{P}_t/dt}{1 - \mathbf{p}_t}. \quad (\text{OA.3.8})$$

Substituting $d\mathcal{P}_t/dt = A(1 - \mathbf{p}_t) - \lambda_H \mathbf{p}_t$ into the belief evolution $d\mathbf{p}_t = (1 - \mathbf{p}_t)[\mu - \lambda_H \mathbf{p}_t - d\mathcal{P}_t/dt]dt$ gives $d\mathbf{p}_t = (1 - \mathbf{p}_t)[\mu - A(1 - \mathbf{p}_t)]dt$. Solving this differential equation yields:

$$\mathbf{p}_t = \frac{De^{\mu t} + (A - \mu)}{De^{\mu t} + A}. \quad (\text{OA.3.9})$$

Plugging this back into (OA.3.8) yields the patenting intensity:

$$\frac{d\mathcal{P}_t}{dt} = \frac{(A + \lambda_H)\mu}{De^{\mu t} + A} - \lambda_H.$$

If $D \neq 0$, then $\lim_{t \rightarrow \infty} d\mathcal{P}_t/dt = -\lambda_H$, which contradicts the requirement that the patenting intensity must be non-negative ($d\mathcal{P}_t/dt \geq 0$). Thus, $D = 0$. This implies that $\mathbf{p}_t = (A - \mu)/A \equiv \mathbf{p}^*$ for all $t \geq T_1$.

For this belief to be valid ($\mathbf{p}^* \geq 0$), we require $A \geq \mu$, which algebraically simplifies to $\rho \leq \rho_M$. Additionally, because the silent phase lasts until T_1 , firms do not patent before T_1 . The belief $\mathbf{p}_t$ must therefore be strictly bounded below the no-patenting asymptote μ/λ_H . Thus, $\mathbf{p}^* < \mu/\lambda_H$, which simplifies to $\rho > \rho_D$. This proves (i). Finally, with $D = 0$, $d\mathcal{P}_t/dt = \mu - \lambda_H \mathbf{p}^*$. Calculating V_t^n with constant $\mathbf{p}_s = \mathbf{p}^*$ and constant patenting intensity confirms that $V_t^n = V_{11} + (1 - \beta \mathbf{p}^*)t^*$, yielding $\Phi(t) = 0$ for all $t \geq T_1$, proving (ii).

(c) **Case $T_2 < \infty$ (Immediate Finish):** After T_2 , the immediate phase begins and $\Phi(t)$ must be strictly decreasing. Thus, $d\Phi(t) < 0$ for $t > T_2$. Because any firm that discovers the technology after T_2 patents it instantly, the stock of unpatented discoveries is zero: $\mathbf{p}_t = 0$. Consequently, the patenting intensity exactly equals the discovery rate: $d\mathcal{P}_t = \mu dt$. Evaluating the evolution equation for $t > T_2$ yields:

$$d\Phi(t) = K_t V_{11} [\lambda_H - \rho(\lambda_H + \beta\mu + 2(1 - \beta)\mu)] dt < 0.$$

Rearranging this strict inequality yields $\lambda_H < \rho(\lambda_H + \mu(2 - \beta))$, which simplifies to $\rho > \rho_M$. $\square$

We are now ready to prove Theorem 2 using this lemma.

Proof of Theorem 2. We prove the theorem by combining the necessary conditions established in Lemma OA.3.8 with the constructive existence of the corresponding patenting strategies.

Part (a): $\rho > \rho_M$ By Lemma OA.3.8 (a), $T_1 = \infty$ requires $\rho \leq \rho_D$, which is violated here. Thus, $T_1 < \infty$. By Lemma OA.3.8 (b), $T_2 = \infty$ requires $\rho \in (\rho_D, \rho_M]$, which is also violated here. Thus, $T_2 < \infty$.

Now, consider any gradual strategy where $0 \leq T_1 < T_2 < \infty$. During the mixing phase $t \in (T_1, T_2)$, the indifference condition requires $d\Phi_t = 0$. Furthermore, because all accumulated discoveries must be patented by T_2 , the unpatented discovery pool is fully depleted, meaning $\lim_{t \rightarrow T_2^-} \mathbf{p}_t = 0$. Using the belief evolution equation (OA.3.9) evaluated at T_1 and taking the limit as $t \rightarrow T_2^-$, we can solve for the integration constant D in two ways:

$$D = \left(\frac{\mu}{1 - \mathbf{p}_{T_1}} - A \right) e^{-\mu T_1} = (\mu - A) e^{-\mu T_2}.$$

Because $\mathbf{p}_{T_1} \geq 0$, we have $\frac{\mu}{1 - \mathbf{p}_{T_1}} \geq \mu$, which implies $\frac{\mu}{1 - \mathbf{p}_{T_1}} - A \geq \mu - A$. Additionally, the condition $\rho > \rho_M$ algebraically simplifies to $A < \mu$, meaning $\mu - A > 0$. Therefore, the right-hand side is strictly positive. However, because $T_2 > T_1$, we also have $e^{-\mu T_1} > e^{-\mu T_2}$.

Multiplying these strict inequalities yields:

$$\left(\frac{\mu}{1 - \mathbf{p}_{T_1}} - A \right) e^{-\mu T_1} > (\mu - A) e^{-\mu T_2},$$

which contradicts the equality for D established above. Therefore, no equilibrium can feature $T_1 < T_2$, leaving only $T_1 = T_2$. It remains to exclude a silent-then-immediate switch at a positive time, $0 < T_1 = T_2 < \infty$. During the silent phase $[0, T_1)$ no patents are filed, so the pool of unpatented discoveries grows strictly from zero and $\lim_{t \rightarrow T_1^-} \mathbf{p}_t > 0$. Entering the immediate phase at T_1 would require this entire accumulated pool to be filed at the single instant T_1 —an atom in the patenting distribution—which by Lemma OA.3.6 cannot occur in a symmetric research equilibrium. Hence $\lim_{t \rightarrow T_1^-} \mathbf{p}_t = 0$, forcing $T_1 = 0$. The only remaining possibility is thus $T_1 = T_2 = 0$, the immediate patenting strategy, which by Theorem 1 is mutually optimal when $\rho > \rho_M$; it is therefore the unique equilibrium.

Part (b): $\rho_M > \rho > \rho_D$ By Lemma OA.3.8 (c), $T_2 < \infty$ requires $\rho > \rho_M$, which is violated here. Thus, $T_2 = \infty$. By Lemma OA.3.8 (a), $T_1 = \infty$ requires $\rho \leq \rho_D$, which is also violated. Thus, any equilibrium must feature $0 \leq T_1 < \infty$ and $T_2 = \infty$ (Delayed Mixing). To prove existence and uniqueness of T_1 , note that during the silent phase $[0, T_1]$, $d\mathcal{P}_t = 0$. The belief $\mathbf{p}_t$ strictly increases from 0 and asymptotically approaches μ/λ_H . By Lemma OA.3.8(b), the mixing phase requires the belief to be exactly $\mathbf{p}^* = 1 - \frac{2\mu(1-\beta)\rho}{\lambda_H(1-\rho) - \mu\beta\rho}$. Because $\rho \in (\rho_D, \rho_M)$, simple algebra confirms that $\mathbf{p}^* \in (0, \mu/\lambda_H)$. Because $\mathbf{p}_t$ is strictly monotonic, there exists a unique, finite time $T_1 > 0$ such that $\mathbf{p}_{T_1} = \mathbf{p}^*$. For $t \geq T_1$, the firm randomizes such that the patenting intensity exactly sustains $\mathbf{p}_t = \mathbf{p}^*$, holding $\Phi(t) = 0$. This explicitly constructs the unique delayed mixing equilibrium.

Part (c): $\rho < \rho_D$ By Lemma OA.3.8 (b) and (c), any equilibrium with $T_2 < \infty$ requires $\rho > \rho_M$, and any equilibrium with a mixing phase extending to infinity ($T_1 < \infty, T_2 = \infty$) requires $\rho > \rho_D$. Since $\rho < \rho_D < \rho_M$, neither is possible. Thus, the only possible equilibrium structure is full concealment ($T_1 = T_2 = \infty$). To verify existence, if both firms fully conceal, $d\mathcal{P}_t = 0$ for all t . As derived in the proof of Lemma OA.3.8 (a), under zero patenting, $\Phi(t) < 0$ globally if and only if $\rho \leq \rho_D$. Since $\Phi(t)$ remains strictly negative, full concealment is the

unique best response. □

OA.3.4 Proofs of Lemmas

Proof of Lemma OA.3.1. Define

$$M_t := e^{-\mu t},$$

$$L_t := \int_0^t \mu M_s e^{-\lambda_H(t-s)} (1 - G_s(t)) ds,$$

where M_t is the probability that the firm has not discovered the new technology. L_t is the probability that the new technology is discovered but the product has not yet been developed and the patent is not filed yet. Then, $\mathbf{p}_t$ is defined by $L_t/(M_t + L_t)$.

Observe that

$$dM_t = -\mu \cdot (1 - \mathbf{p}_t) dt \cdot (M_t + L_t), \tag{OA.3.10}$$

$$\begin{aligned} dL_t &= \mu M_t (1 - G_t(t)) dt - \lambda_H L_t dt - \int_0^t \mu M_s e^{-\lambda_H(t-s)} dG_s(t) ds \\ &= [\mu(1 - \mathbf{p}_t) dt - \lambda_H \mathbf{p}_t dt - d\mathcal{P}_t] \cdot (M_t + L_t). \end{aligned} \tag{OA.3.11}$$

By differentiating $\mathbf{p}_t/(1 - \mathbf{p}_t) = L_t/M_t$,

$$\frac{d\mathbf{p}_t}{(1 - \mathbf{p}_t)^2} = \frac{\mathbf{p}_t}{1 - \mathbf{p}_t} \left[\frac{dL_t}{L_t} - \frac{dM_t}{M_t} \right] = \frac{\mathbf{p}_t}{1 - \mathbf{p}_t} \left[\frac{\mu(1 - \mathbf{p}_t) dt - \lambda_H \mathbf{p}_t dt - d\mathcal{P}_t}{\mathbf{p}_t} + \mu dt \right].$$

By rearranging this equation, we obtain (OA.3.1). □

Proof of Lemma OA.3.2. By definition, V_t^n represents Firm A's expected continuation payoff from time t onward under the full concealment policy ($\tilde{G}_t(s) = 0$ for all $s \geq t$).

Under this policy, Firm A receives payoffs only through two events: either (i) Firm A develops the product first (net of cost), yielding $(\lambda_H \Pi - c)dt$; or (ii) Firm B files a patent, yielding $(V_{11} - (1 - \beta) \cdot l^*)d\mathcal{P}_t$. The term K_t/K_τ serves as the effective discount factor, representing the conditional probability that neither firm has ended the race by time t , given

that it was ongoing at time τ . Integrating these flow payoffs yields:

$$V_t^n = \int_{\tau}^{\infty} (\lambda_H \Pi - c) \cdot \frac{K_t}{K_{\tau}} dt + \int_{\tau}^{\infty} (V_{11} - (1 - \beta) \cdot l^*) \cdot \frac{K_t}{K_{\tau}} d\mathcal{P}_t,$$

which is equivalent to (OA.3.4).

Note that

$$\int_t^{\infty} d\mathcal{F}_s = (\lambda_H \Pi - c) \cdot \int_t^{\infty} K_s ds + (V_{11} - (1 - \beta) \cdot l^*) \cdot \int_t^{\infty} K_s d\mathcal{P}_s.$$

The first term is finite since $0 \leq K_s \leq e^{-\lambda_H s}$.

For the second term, observe that $K_s d\mathcal{P}_s$ represents the *discounted* probability density or mass of the firm patenting at time s . Thus, the integral is bounded by the total probability of patenting:

$$\int_t^{\infty} K_s d\mathcal{P}_s \leq \int_t^{\infty} 1 \cdot d\Pr(\text{The firm patents at } s) \leq 1.$$

Since both integrals are finite and $K_t > 0$, the value V_t^n is finite. $\square$

Proof of Lemma OA.3.3. Note that Firm A's expected payoff can be written as follows:

$$\begin{aligned} V_{\tau}(\tilde{G}_{\tau}, G) := & \tilde{G}_{\tau}(\tau) \cdot (V_{11} + (1 - \beta \mathbf{p}_{\tau}) \cdot l^*) + \int_{\tau}^{\infty} (V_{11} + (1 - \beta \mathbf{p}_t) \cdot l^*) \cdot \frac{K_t}{K_{\tau}} d\tilde{G}_{\tau}(t) \\ & + \int_{\tau}^{\infty} (\lambda_H \Pi - c) \cdot \frac{K_t}{K_{\tau}} \cdot (1 - \tilde{G}_{\tau}(t)) dt \\ & + \int_{\tau}^{\infty} (V_{11} - (1 - \beta) \cdot l^*) \cdot \frac{K_t}{K_{\tau}} \cdot (1 - \tilde{G}_{\tau}(t)) d\mathcal{P}_t. \end{aligned}$$

The first two terms correspond to the payoffs when Firm A patents. The third term corresponds to the case where Firm A develops the product first, net of cost. The last term is the payoff when Firm B patents.

Observe that:

$$\begin{aligned}
V_\tau(\tilde{G}_\tau, G) &= \tilde{G}_\tau(\tau) \cdot (V_{11} + (1 - \beta \mathbf{p}_\tau) \cdot l^*) + \frac{1}{K_\tau} \int_\tau^\infty (1 - \tilde{G}_\tau(t)) d\mathcal{F}_t \\
&\quad + \frac{1}{K_\tau} \int_\tau^\infty (V_{11} + (1 - \beta \mathbf{p}_t) \cdot l^*) \cdot K_t d\tilde{G}_\tau(t) \\
&= \tilde{G}_\tau(\tau) \cdot (V_{11} + (1 - \beta \mathbf{p}_\tau) \cdot l^*) + (1 - \tilde{G}_\tau(\tau)) \cdot V_\tau^n \\
&\quad - \frac{1}{K_\tau} \int_\tau^\infty \int_\tau^t d\tilde{G}_\tau(s) d\mathcal{F}_t + \frac{1}{K_\tau} \int_\tau^\infty (V_{11} + (1 - \beta \mathbf{p}_t) \cdot l^*) \cdot K_t d\tilde{G}_\tau(t) \\
&= V_\tau^n + \tilde{G}_\tau(\tau) \cdot \frac{\Phi(\tau)}{K_\tau} - \frac{1}{K_\tau} \int_\tau^\infty \int_t^\infty d\mathcal{F}_s d\tilde{G}_\tau(t) \\
&\quad + \frac{1}{K_\tau} \int_\tau^\infty (V_{11} + (1 - \beta \mathbf{p}_t) \cdot l^*) \cdot K_t d\tilde{G}_\tau(t) \\
&= V_\tau^n + \frac{1}{K_\tau} \left[\tilde{G}_\tau(\tau) \cdot \Phi(\tau) + \int_\tau^\infty \Phi(t) d\tilde{G}_\tau(t) \right]. \tag{OA.3.12}
\end{aligned}$$

Thus, (OA.3.6) holds. $\square$

Proof of Lemma OA.3.4. If $\sup_{t \geq \tau} \Phi(t) < 0$, the above expression is maximized when $\tilde{G}_\tau(\tau) = 0$ and $d\tilde{G}_\tau(t) = 0$ for all $t \geq \tau$. Thus, full concealment is optimal.

Next, consider the support of the optimal policy G_τ^* . Note that the objective function is linear in the probability measure $d\tilde{G}_\tau$. Consequently, the expected payoff is maximized if and only if the probability measure is concentrated entirely on the set of times that maximize the integrand $\Phi(t)$. Thus, if T is in the support of dG_τ^* , it must be that $T \in \arg \max_{t \geq \tau} \Phi(t)$.

Furthermore, if $\Phi(T) < 0$ for such a T , then the maximum value of the integrand is negative. In this case, as shown above, the optimal policy is full concealment (zero measure everywhere), which implies the support is empty (or does not contain T). Hence, for T to be in the support, we must have $\Phi(T) \geq 0$. $\square$

Proof of Lemma OA.3.5. Note that $\Phi(t) = (V_{11} + (1 - \beta \mathbf{p}_t) \cdot l^*) \cdot K_t - \int_t^\infty d\mathcal{F}_s$. By taking a derivative, we have

$$d\Phi = -\beta l^* K_t d\mathbf{p}_t + (V_{11} + (1 - \beta \mathbf{p}_t) \cdot l^*) dK_t + d\mathcal{F}_t.$$

Using (OA.3.10) and (OA.3.11), we can derive that

$$dK_t = -K_t (\lambda_H(1 + \mathbf{p}_t)dt + d\mathcal{P}_t). \quad (\text{OA.3.13})$$

Using this together with Lemma OA.3.1 and (OA.3.5), we can rewrite $d\Phi$ as follows:

$$\begin{aligned} d\Phi = & -\beta l^* K_t (1 - \mathbf{p}_t) [(\mu - \lambda_H \mathbf{p}_t)dt - d\mathcal{P}_t] \\ & - (V_{11} + (1 - \beta \mathbf{p}_t)l^*) K_t [\lambda_H(1 + \mathbf{p}_t)dt + d\mathcal{P}_t] \\ & + K_t [2\lambda_H V_{11}dt + (V_{11} - (1 - \beta)l^*)d\mathcal{P}_t]. \end{aligned}$$

By rearranging this equation, we obtain (OA.3.7). $\square$

Proof of Lemma OA.3.6. Suppose that there is a discrete downward jump of Φ at time T . This can happen only when firms patent with a mass at time T , which induces a discontinuous downward jump of Φ at time T :

$$\Delta\Phi(T) = -K_T[2(1 - \beta)l^*]\Delta\mathcal{P}_T \leq 0.$$

Since Φ is right-continuous from the right-continuity of G , $\Phi(T)$ cannot be the maximum of Φ , which contradicts that there is a mass of patenting at time T from Lemma OA.3.4. $\square$

Proof of Lemma OA.3.7. Suppose that there exists $S \in (T, T')$ such that $\Phi(S) < \Phi(T) = \Phi(T')$. By Lemma OA.3.6, Φ is continuous. Thus, there exists an interval $[S_1, S_2] \subset (T, T')$ such that $\Phi(t) < \Phi(T)$ for all $t \in [S_1, S_2]$, and $d\Phi(S_1) < 0$ and $d\Phi(S_2) > 0$.

Note that for any $t \in [S_1, S_2]$, since $\Phi(t) < \Phi(T)$, firms do not patent at time t from Lemma OA.3.4. Thus, $d\mathcal{P}(t) = 0$ for all $t \in [S_1, S_2]$ and $\mathbf{p}_{S_1} < \mathbf{p}_{S_2}$. Then, $d\Phi(S_1) < 0$ implies from (OA.3.7) that

$$\rho > \frac{\lambda_H}{\lambda_H + \beta\mu + 2\lambda_H(1 - \beta)\frac{\mathbf{p}_{S_1}}{1 - \mathbf{p}_{S_1}}} > \frac{\lambda_H}{\lambda_H + \beta\mu + 2\lambda_H(1 - \beta)\frac{\mathbf{p}_{S_2}}{1 - \mathbf{p}_{S_2}}}.$$

This implies that $d\Phi(S_2) < 0$ from (OA.3.7), which contradicts the assumption that $d\Phi(S_2) > 0$. $\square$